

Sunilraj D

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Education

Amrita Vishwa Vidyapeetham

Chennai, India

B.Tech – Computer Science & Engineering (Artificial Intelligence)

2023 – 2027

– CGPA: 7.64 / 10 | Relevant Coursework: Deep Learning, Computer Vision, NLP, Data Structures & Algorithms

Skills

Languages: Python, Java, C, SQL

ML / DL: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Hugging Face Transformers

NLP: BERT, Sentence Transformers, TF-IDF, LLM Prompt Engineering, Embeddings

Computer Vision: CNNs, UNet, Image Segmentation, Transfer Learning, OpenCV

Tools & Deployment: Linux, Streamlit, Flask, Pandas, NumPy, Matplotlib, Seaborn, Git, Jupyter

Certifications & Achievements

NPTEL: Python for Data Science — **Top 2%**, Score 83% | C Programming — Score 78%

Coursera: Data Structures & Algorithms

Teachnook & IIT Roorkee: Data Science Certification

Experience

Warehouse Intelligence Systems Intern

Chennai, India

TAFE – Tractors and Farm Equipment Limited

Feb 2026 – May 2026

- Built an end-to-end intelligent warehouse management system in **Python + Streamlit**, replacing manual Excel-based inventory tracking across the engineering stores department.
- Designed a dynamic weighted scoring system — $\text{score} = (\text{usage} \times \text{weight}) - (\text{days_inactive} \times \text{decay})$ — with movement multipliers (Fast 1.5 $\times$, Slow 1.0 $\times$, Rare 0.5 $\times$) to automate shelf placement recommendations.
- Implemented **K-Means clustering (k=3)** on 6-month usage logs to classify parts into Fast/Slow/Rare movement categories, validated using Elbow method and Silhouette score.

Projects

Zero-Shot EV Energy Prediction via NLP Knowledge Transfer *BERT, Sentence Transformers, XGBoost, LSTM, PCA, TF-IDF*

- Developed a zero-shot prediction system that estimates EV energy consumption from owner manuals alone — **no historical driving data required** — achieving only **2.6% error** on unseen vehicles.
- Trained LSTM and XGBoost models on 5,000 structured EV sensor records achieving a baseline **R² of 0.9378**; applied BERT-based feature weighting and PCA embeddings to push **R² to 0.9394**.
- Extracted domain knowledge from PDF manuals using TF-IDF and Sentence Transformers to map textual specs to physical features, enabling cross-modal transfer for new vehicle models.

Attention-Enhanced UNet for Lung Tumor Segmentation

PyTorch, UNet, Attention Gates, SE Blocks, Hybrid Loss

- Built an Attention Gate UNet for lung tumor segmentation on **14K+ CT slices**, achieving a **Dice score of 0.9456**, Precision of 0.9523, and Recall of 0.9603.
- Designed a custom **Hybrid Loss** (Dice + Focal + Boundary) to tackle severe class imbalance and sharpen segmentation boundaries on tiny tumor regions.
- Integrated SE (Squeeze-and-Excitation) blocks for channel-wise feature recalibration, improving localization accuracy in ambiguous scan regions.

Hybrid CNN for Alzheimer's MRI Classification

TensorFlow, CNN, PCA, SVM, XGBoost, Random Forest, Stacking

- Designed a two-stage hybrid pipeline: CNN for deep feature extraction from MRI scans, followed by a stacking ensemble (SVM + XGBoost + Random Forest) for 4-class impairment classification.
- Achieved **90.77% accuracy** and **ROC-AUC of 0.9826** on the Alzheimer's MRI dataset; reduced inference time by replacing the CNN head with lightweight ML classifiers.
- Published as a research paper; contributed model architecture, training pipeline, and full evaluation framework.

LLM-Powered Document Structuring Pipeline

Python, Gemini API, PyPDF2, Prompt Engineering, Pandas

- Built a fully automated pipeline to extract structured data from unstructured PDFs into JSON/Excel using **Google Gemini API** with custom prompt engineering.
- Designed a normalization layer to handle missing, inconsistent, and malformed fields; generated **60+ structured records per document** with reliable, context-aware outputs.